# Computer Networks 2 - Exam March 2024

Maximum points: 36p
Preliminary grading: Pass (3): 19p, (4): 26p, (5): 32p

NAME / ANONYMOUS NUMBER:

All answers must be in English. Start a new page for each problem.

Read the questions carefully and answer what is asked for! Your task is to convince the teachers who are grading the exam that you have a certain level of knowledge. The best way to do this are concise, comprehensible and clear answers to the questions asked – and nothing else.
In some of the problems, you may not be given all the details needed to answer the problem. In such cases, you need to make realistic assumptions that are explicitly stated and motivated. Omitting an assumption or not motivating your assumptions will give you less points.

Also, remember to **write in a clear, readable handwriting**.

*Good luck with the exam!*

| Problem Number | Solved | Points |
|---|---|---|
| 1 | | |
| 2 | | |
| 3 | | |
| 4 | | |
| 5 | | |
| 6 | | |
| Sum | | |
| Bonus Points | | |
| Grade | | |

Please set a X in the Solved column for each problem you have worked on

# Problem 1: Short Questions (10p)

In this problem you can give short answers without motivation, unless stated differently.

a. You connect to a new server using QUIC. Can your QUIC client perform a zero-RTT connection establishment to that server?
   A: No, since it is a new server.

b. Describe briefly an example of a scenario where a connection migration of a QUIC connection is needed.
   A: When a change in the access network occurs.

c. Assume a random variable X that follows a geometric distribution with p=0.4. Which value is higher: P(X=3) or P(X=4)?
   A: $P(X = i) = (1 - p)^{i-1}p$. Hence, $P(X = 3) = 0.6^2 \times 0.4$ and $(X = 4) = 0.6^3 \times 0.4$. Therefore, P(X=3) is higher.

d. The binomial distribution is a discrete probability distribution of the number of successes of a series of independent Bernoulli experiments. Is this correct?
   A: Yes.

e. Mention one characteristic of the Kubernetes networking model.
   A: Container addressability, topological simplicity, and integration (see slide 41 of Simone's guest lecture).

f. What is the main benefit of passive network measurements over active network measurements?
   A: No overhead (see slide 33 of Andreas guest lecture).

g. Mention two problems with the host-centric view of today's Internet (max one sentence per problem).
   A: Information is tied to its location, information distribution is inefficient, mobility and disruptions become difficult to handle, cannot trust an object copy received from an untrusted server (see slide 7 of Bengt's guest lecture).

h. What is the difference between transmission and queueing delay?
   A: Transmission delay is the amount of time required to send all the bits of a packet into the transmission medium. Queuing delay is the amount of time a packet waits in line before being transmitted over a link or processed by a routing device.

i. What is one of the main performance bottlenecks of today's networks?
   A: Latency (see slide 32 of Bengt's guest lecture).

j. Give three examples of network performance metrics.
   A: Availability, one-way delay (s), delay variation (s), round-trip time (s), packet loss (%), error ratio (%), throughput (bits/s), available capacity (bits/s).

## Problem 2: Power-saving (3p)

Given the increase in cost for power, servers are typically only on when being used, and turned off whenever they are not in use. Assume now the following power-aware algorithm: when a job arrives, a fresh server is turned on (assume zero setup cost). As soon as the job completes service, the server is turned off. Also assume that there is always a server available for every job, so there is no queueing. Assume that when a server is on, it consumes power at a rate of P = 120 Watts. Assume λ = 20 jobs arrive per second, and that the service requirement of jobs is exponentially-distributed with mean 10 seconds.

Derive the average rate at which power is used in the system.

## Solution:

Use Little's Law, which implies that $E[N] = \lambda E[T]$. Since jobs are exponentially-distributed with mean 10 seconds, this implies that $E[T] = 10$ (1p until here).

Now, we calculate the average number of jobs from Little's law as: $E[N] = 20 \times 10 = 200$ (1p here). Since the server consumes power at a rate of P = 120 Watts, we obtain the average power consumption rate as: $200 \times 120 = 24000$ Watts (1p here).

# Problem 3: Queueing (7p)

Assume an M/M/K/2K server system, i.e., a networking system with K servers and some queueing capacity. The arrival rate is $\lambda$, the service rate of each server is $\mu$, and the packet loss due to full buffer is $P_{block}$. For this system, we can define the throughput as $\lambda(1 - P_{block})$.

Answer the following questions, and justify each answer (1-5 lines max for each question). Note that you do not need to perform any derivations using the formulas (unless you prefer).

a) What is the queueing capacity of the system? (1p)
b) Assume that you increase the number of servers to 2K, while everything else stays the same, including the queueing capacity. Does the throughput of the server system increase or decrease or stay the same? (2p)
c) Continue with the assumption from question b. Does the service time E[S] for the jobs that receive service increase, decrease or stay the same? (1p)
d) Assume now that you keep the servers to K, but you duplicate the queueing capacity. Does the throughput of the server increase or decrease or stay the same? (1p)
e) Continue with the assumption from question d. Does the service time E[S] for the jobs that receive service increase, decrease or stay the same? (1p)
f) Continue with the assumption from question d. Does the mean response time E[W] for the jobs that receive service increase, decrease or stay the same? (1p)

## Solution:

a) The queueing capacity is K because the system has K servers and supports 2K customers, thus implying that K customers can be queued.
b) This is now an M/M/2K/3K system. Since there are more servers in the system, they drain the queue of size K faster than before. This means that the probability to find the queue full becomes lower, i.e., the blocking probability is lower. Since the throughput of a system with blocking is $(1 - P_{block})$ the throughput now increases.
c) The service time is affected only by the individual server speed ($\mu$) and not the number of servers. It remains the same.
d) This is an M/M/K/3K system. Since there is a bigger queue now, and the total server capacity remains the same, the packet loss probability decreases, and throughput increases because more arriving packets are eventually served.
e) The service time is affected only by the individual server speed and not the queue size. It remains the same.
f) The service time for each job is the same, but the queueing time now can be larger (since the queue is double the size but the speed it gets drained remains the same).

# Problem 4: Markov Chains (7p)

Consider a network router that operates in three states: normal, congested, and failed.
In the normal state:

- It remains in the normal state with a probability of 0.9.
- It transitions to the congested state with a probability of 0.05.
- It transitions to the failed state with a probability of 0.05.

In the congested state:

- It stays congested with a probability of 0.5.
- It returns to the normal state with a probability of 0.1.
- It transitions to the failed state with a probability of 0.4.

In the failed state:

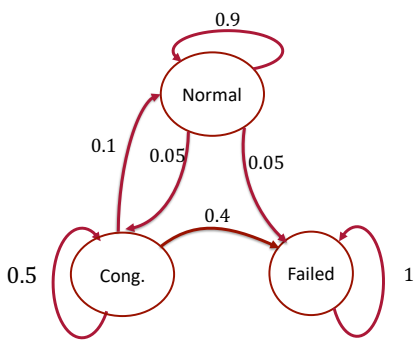
- It remains failed with a probability of 1.
- It transitions to the normal state with a probability of 0.
- It transitions to the congested state with a probability of 0.

Now, answer the following questions:

a) Derive the state transition matrix and draw the Markov Chain. (1p)
b) Is the Markov Chain irreducible? Motivate your answer. If not, explain what change(s) are necessary to make it irreducible. (2p)
c) A network operator reconfigures the router and the probabilities in the failed state change. Now, the network remains failed with a probability of 0.2, transitions to the normal state with probability 0.6, or transitions to the congested state with probability 0.2. Derive the new state transition matrix and draw the new Markov Chain. (1p)
d) Assume the new network router configuration starts in the normal state. Calculate the probability vector after three-time units. (2p)
e) Write a pseudocode to compute the stationary distribution of the new Markov Chain. (1p)

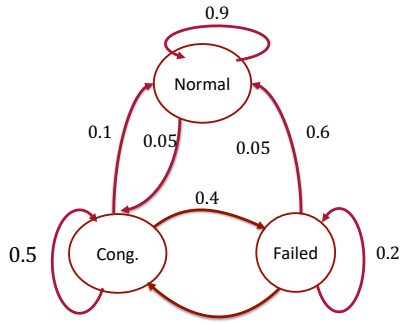
## Solution:

a) The state transition matrix is: $P = \begin{pmatrix} 0.9 & 0.05 & 0.05 \\ 0.1 & 0.5 & 0.4 \\ 0 & 0 & 1 \end{pmatrix}$. The Markov Chain is as follows:




b) The Markov Chain is reducible because the Failed state is a recurrent state. Thus, once the Markov Chain arrives at the Failed state, it never leaves. To change this, we need to change the network and reconfigure the router to avoid 100% failed probability.

c) The new state transition matrix is: $P = \begin{pmatrix} 0.9 & 0.05 & 0.05 \\ 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \end{pmatrix}$. The new Markov Chain is as follows:




d) For this, consider that $\pi^0 = (1\ 0\ 0)$. Since the Markov Chain is irreducible, aperiodic, and homogeneous, the Normal equations hold. Therefore, the probability vector after three-time units is $\pi^{(3)} = \pi^{(0)}P^3$. After the calculations, we obtain that:

$P = \begin{pmatrix} 0.9 & 0.05 & 0.05 \\ 0.1 & 0.5 & 0.4 \\ 0.6 & 0.2 & 0.2 \end{pmatrix}$, $P^2 = \begin{pmatrix} \mathbf{0.845} & \mathbf{0.08} & \mathbf{0.075} \\ 0.38 & 0.335 & 0.285 \\ 0.68 & 0.17 & 0.15 \end{pmatrix}$,

$P^3 = \begin{pmatrix} \mathbf{0.8135} & \mathbf{0.09725} & \mathbf{0.08925} \\ 0.5465 & 0.2435 & 0.21 \\ 0.719 & 0.149 & 0.132 \end{pmatrix}$. Therefore, $\pi^{(3)} = (0.8135\ \ 0.09725\ 0.08925)$.

e) See the image below:

```
# For the stationary distribution, do this for n times
n_station = 100
staPiP =  pi
for idx in range(n_station):
    staPiP = staPiP @ P

print("Stationary distribution:", staPiP)
```

# Problem 5: Channel Capacity (3p)

Consider the Channel Capacity Theorem for a continuous communication channel with bandwidth W Hz, perturbed by the constant additive white Gaussian noise of power spectral density $N_0$, and the average transmitted power P.

Answer the following questions:

a) Is there any limit to the capacity of such a channel if you increase its signal-to-noise ratio $\frac{P}{N_0 W}$ without limit? If so, what is that limit? (1p)

b) Consider the situation in question a. What are the impacts of such increase in the signal-to-noise ratio without limits? Motivate your answer. (1p)

c) If you can increase the bandwidth W without limits and while fixing P, how will the Channel Capacity be affected? Is there a limit? (1p)

## Solution:

a) The capacity of such a channel, in bits per second, is $C = W \log_2\left(1 + \frac{P}{N_0 W}\right)$. Increasing the quantity $\frac{P}{N_0 W}$ inside the logarithm without bounds causes the capacity to increase monotonically and without bounds.

b) Such increase in the signal-to-noise-ratio is not possible because it implies in the increase of the transmitted power without limits. This leads to severe health issues and is strictly regulated by law. Therefore, the transmitted power cannot increase without limits.

c) Increasing the bandwidth W alone causes a monotonic increase in capacity, but only up to an asymptotic limit. That limit can be evaluated by observing that in the limit of small $\alpha$, the quantity $\log_e(1 + \alpha)$ approaches α. In this case, setting $\alpha = P$ and defining $W' = N_0 W$, if we let $W'$ become arbitrarily large, the capacity C becomes:

$C = \frac{W'}{N_0}\log_2\left(1 + \frac{\alpha}{W'}\right)$, $C_{\lim} = \lim_{W' \to \infty} \frac{W'}{N_0}\log_2\left(1 + \frac{\alpha}{W'}\right) = \frac{\alpha}{N_0}\log_2(e) = \frac{P}{N_0}\log_2(e)$.

Thus, there are vanishing returns from endless increasing in bandwidth, unlike the unlimited returns enjoyed from improvement in signal-to-noise ratio.

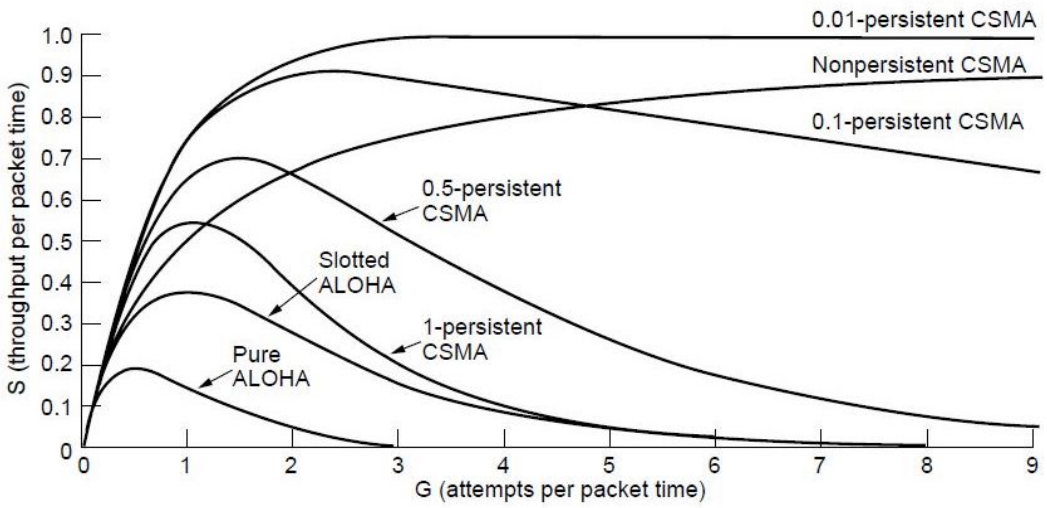
## Problem 6: ALOHA and CSMA (6p)

One important difference between ALOHA and CSMA is carrier sensing, which means that CSMA listens for ongoing transmissions to terminate before starting a transmission. Transmissions after the transmission medium becomes idle can be done more or less "aggressive" based on a parameter p (p-persistent CSMA) as follows:

> "When the transmitting node is ready to transmit data, it senses the transmission medium for idle or busy. If idle, then it transmits immediately. If busy, then it senses the transmission medium continuously until it becomes idle, then transmits with probability p. If the node does not transmit (the probability of this event is 1-p), it waits until the next available time slot. If the transmission medium is not busy, it transmits again with the same probability p. This probabilistic hold-off repeats until the frame is finally transmitted or when the medium is found to become busy again (i.e. some other node has already started transmitting). In the latter case the node repeats the whole logic cycle (which started with sensing the transmission medium for idle or busy) again."

The following figure compares ALOHA and CSMA. For this problem, you can consider S on the y-axis to be the efficiency of the protocols, and you should ignore G on the x-axis. In other words, consider the maximum efficiency of each protocol.

a. Relating to probability theory, argue why Slotted ALOHA achieves higher efficiency than Pure ALOHA (nodes can start to transmit at any time). That is, for both protocols explain when a collision occurs and relate the probabilities to each other. You do not have to calculate the maximum efficiency. (2p)
b. Argue why 0.01-persistent CSMA achieves higher efficiency than 1-persistent CSMA. (2p)
c. What is the drawback of CSMA with lower persistency p? Discuss the trade-off between efficiency and this drawback. (2p)




## Solution:

a) The essence of why Slotted ALOHA is more efficient than Pure ALOHA is that by reducing the vulnerability period by half, Slotted ALOHA decreases the probability of

collisions, leading to higher throughput and efficiency. For the calculations, see slides 29 and 32 for the probabilities.

b) 0.01-persistent CSMA achieves higher efficiency than 1-persistent CSMA by significantly reducing the probability of collisions. The efficiency gains come from the decreased likelihood of needing to retransmit frames, which saves bandwidth and reduces the overall network delay.

c) The trade-off here is between achieving higher network efficiency by reducing collisions and the potential downside of increased delay and underutilization of network capacity. With lower persistency, the efficiency gains are realized through reduced collision rates, which means fewer retransmissions, less overhead, and potentially higher throughput for the network as a whole. The main drawback is the potential for increased latency and reduced network throughput, particularly in low-traffic scenarios, as nodes delay their transmissions even when the channel is idle.

   The difference in approach essentially balances the risk of collision against the desire for immediate access to the channel. Lower persistency aims to minimize collisions at the cost of potentially increasing the average time to successfully transmit a frame.